

HDOC - Day 14

Unique Digits and Digit Frequency

Algorithm

1. Start. 2. Read positive integer n. 3. Display menu. 4. Initialize frequency values for digits 0-9 to zero. 5. For option 1, extract each digit, increment its frequency and mark the number Not Unique if any frequency exceeds 1; otherwise display Unique. 6. For option 2, count every digit and display frequencies from 0 through 9. 7. Repeat until Exit. 8. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	12345; choice 1	Unique	Unique	Pass
2	12234; choice 1	Not Unique	Not Unique	Pass
3	112333; choice 2	f1=2,f2=1,f3=3; others 0	Matches expected	Pass

C Program

```
#include <stdio.h>
int main() {
    int n,choice,temp,d,i,freq[10],unique;
    do { printf("Enter a positive integer: "); scanf("%d",&n); } while(n<=0);
    do {
        printf("\n1. Check whether all digits are unique\n");
        printf("2. Frequency of digits 0 to 9\n3. Exit\nEnter choice: ");
        scanf("%d",&choice);
        switch(choice) {
            case 1:
                for(i=0;i<10;i++) freq[i]=0;
                temp=n; unique=1;
                while(temp>0) {
                    d=temp%10; freq[d]++;
                    if(freq[d]>1) unique=0;
                    temp/=10;
                }
                if(unique) printf("Unique\n"); else printf("Not Unique\n");
                break;
            case 2:
                for(i=0;i<10;i++) freq[i]=0;
                temp=n;
                while(temp>0) { d=temp%10; freq[d]++; temp/=10; }
                for(i=0;i<10;i++) printf("freq_%d = %d\n",i,freq[i]);
                break;
            case 3: break;
            default: printf("Invalid choice.\n");
        }
    } while(choice!=3);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day14.c -o day14
puppie@Anvikas-MacBook-Air Desktop % ./day14
Enter a positive integer: 12345
Enter choice: 1
Unique
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day14
Enter a positive integer: 12234
Enter choice: 1
Not Unique
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day14
Enter a positive integer: 112333
Enter choice: 2
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

Result: The menu-driven program was compiled, executed and tested successfully against the prepared test cases.